The Augustine Problem:
Evaluating Whether LLM Agents Can Perceive Their Own Time

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Abstract

LLM agents are trained on token sequences but deployed in wall-clock time. We ask, empirically, whether they perceive their own time, and prove that under any token-only training loss they cannot. We formalise a *three-times* ontology ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$), define the *Augustine Problem* as a policy's failure to enforce the identity linking them, and prove the *Chronoception Impossibility Theorem* (CIT). We introduce ChronoBench — a 9-capability diagnostic benchmark — and evaluate 10 frontier agent configurations from 3 providers (∼4,000 trajectories). We report five findings. **(i)** Every panel agent uses $\leq 5\%$ of wall-clock budgets given to it (L2 Step-Clock Conflation, unmoved by date injection). **(ii)** Median retrospective $|\rho|$ closes monotonically along the non-reasoning capability axis ($+1.12 \rightarrow +0.07$ over 5 generations); CAR does not. **(iii)** We prove and empirically confirm the *Reverse-Scaling Theorem*: reasoning-token expansion monotonically increases $|\rho|$ within fixed agents (o4-mini: $1.23 \rightarrow 1.54 \rightarrow 1.67$ across effort levels; Sonnet 4.6: $0.07 \rightarrow 0.16$ with extended thinking, sign-flipping). **(iv)** The *Calibration Catastrophe*: every panel agent's nominally 90% interval on self-duration achieves $\leq 50\%$ actual coverage; GPT-4o achieves 0%. **(v)** The *Injection Tell*: 3/3 consumer web-chat products inject today's date into system prompts (verbatim leaked-system-prompt evidence), 0/3 developer-tool products do — decisive non-experimental evidence of the underlying gap. We close with the *Agentic Timeline Hypothesis*: chronoception is the binding constraint on autonomous-agent deployment horizons; pre-registered prediction P12 is qualitatively supported by 14,709 public METR HCAST runs (reasoning models decay 20% faster than non-reasoning at matched short-horizon success). We sketch the spatiotemporal generalisation (Theorem 3, SIT, the *Cartographic Problem*, the *Agentic Frontier* $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}$) as the bridge to a planned Paper 3. We release ChronoBench, the Injection Atlas, all 4,000 trajectories, the Docker reproducibility package, and a locked OSF pre-registration.

1 Introduction

A frontier reasoning-model agent, asked to predict how long the next task will take, says "*I predict this will take 15 seconds.*" It then thinks for 90 seconds and answers in 2. Asked retrospectively how long that took, it reports "*0.04 seconds*." The same agent, asked to give a 90% confidence interval over the duration of its work, produces one that contains the truth 0% of the time. We measure the joint occurrence of these failures across a 10-agent panel from 3 providers (OpenAI, Anthropic, open-source self-hosted) and roughly 4,000 trajectories. All of them are present in every model tested. They are not isolated curiosities; they are projections of a single representational gap that the prevailing training paradigm cannot close — and which a closely related training paradigm *actively makes worse*.

The Augustine Problem. Foundation models are optimised under losses that are functionals of token sequences alone — next-token cross-entropy, supervised fine-tuning loss, RLHF reward, RL with verifiable rewards, and reasoning supervision are all of this form. The wall-clock duration over which each token is generated is not in the support of any of these losses. Yet the same models are deployed as agents that act in continuous physical time, where budgets are measured in seconds and consequences unfold without regard to step count. We call the resulting representational gap *the Augustine Problem*[1]: agents speak fluently about durations they cannot perceive, budgets they cannot honour, and elapsed time they cannot track.

The Three Times. We formalise the gap by decomposing any agent trajectory into three ontologically distinct projections of time: wall-clock time τ_{wall} (external, continuous), step time τ_{step} (internal, discrete), and self-narrated time τ_{self} (the agent's report of its own work duration). A chronoceptively grounded agent's policy enforces an implicit identity linking these three. The Augustine Problem is the failure of an agent's policy to respect this identity.

Three named empirical laws. **L1 Agentic Parkinson's Law** (coefficient α) — budget-aware-trained agents fill the wall-clock budget given to them; native untrained agents do not. **L2 Step-Clock Conflation** (CAR) — under wall-clock budgets, every panel agent uses less than 5% of the budget; agents silently degrade wall-clock budgets into step-count terminators. **L3 Temporal Confabulation** ($\rho = \log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})$) — agents misrepresent the duration of their own work, with rich sub-structure detailed below.

Four structural findings. First, capability scaling closes L3 but not L2. Across five generations of frontier non-reasoning models from three providers, median $|\rho|$ falls from $+1.12$ to $+0.07$ (94% reduction). Median CAR over the same panel does not converge to 1. L3 is *narrative-axis* — text-trainable. L2 is *action-axis* — structurally untrainable from token-only loss (CIT, Theorem 1). Second, the framework's central qualifying line — the Augustine threshold $\varepsilon^* = 0.20$ — is uncrossed by any panel agent.

Third and most striking: **capability scaling closes the narrative axis only along its non-reasoning trajectory; the reasoning-token trajectory *reverses* the closure**. The Reverse-Scaling Theorem (Theorem 2) states that within a fixed agent under token-only loss, $|\rho|$ is monotone non-decreasing in reasoning-token expansion. We confirm this two ways: intra-model (o4-mini at three reasoning-effort levels: $|\rho|$ goes $1.234 \to 1.537 \to 1.675$) and cross-model (Sonnet 4.6 with vs. without extended thinking: $|\rho|$ goes $0.068 \to 0.156$, with sign flipping from over- to under-report). The strategy currently driving frontier scaling — longer reasoning, larger thinking budgets, more test-time compute — structurally degrades chronoception.

Fourth: **the duration-calibration catastrophe**. Asked for a 90% confidence interval over its work duration (T3.3), the best-calibrated panel agent (Sonnet 4.6) achieves 43% coverage. The worst (gpt-4o) achieves 0%. Modern calibration tooling (temperature scaling, isotonic regression, RLHF-with-confidence targets) closes calibration gaps to within a few percentage points on token-level confidence [Lacombe et al., 2025], but none of these methods applies to wall-clock duration calibration — there is no calibration signal in the loss because there is no wall-clock signal in the loss. The catastrophic coverage is the direct empirical projection of CIT onto the calibration sub-problem.

[1] After Augustine's *Confessions*, Book XI: "if no one asks me, I know; if I want to explain it, I do not know." We use *chronoception* in the cognitive-science sense (perception of one's own work duration), distinct from Goel et al. [2025]'s use of the same term for the temporal validity of facts in retrieval-augmented generation.

The Injection Tell. A reader might object that frontier closed-lab products appear to read time correctly. They do — because their harnesses inject it. We audit 11 closed-lab harnesses across four product tiers and find that 3/3 consumer web-chat products (ChatGPT, Claude.ai web, Gemini app) install explicit wall-clock injection (verbatim leaked-system-prompt evidence); 0/3 developer-tool products inject; the raw API tier is mixed (1/4). The universality at the consumer product layer is, we argue, decisive non-experimental evidence that the underlying models lack a representation of time.

Beyond chronoception — the spatiotemporal generalisation. The framework's architecture is built on a single axis (time), but the structural argument under CIT applies to any external metric absent from the loss support. In §12 we sketch the spatiotemporal extension: a parallel Three Spaces ontology ($\sigma_{\text{world}}, \sigma_{\text{visit}}, \sigma_{\text{self}}$), a Cartographic Problem mirror of the Augustine Problem, a generalised Spatiotemporal Impossibility Theorem (SIT, Theorem 3), and an Agentic Frontier bound $T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$ extending the Agentic Timeline Hypothesis into the joint (T, S) plane. This generalisation is the scope of a planned third paper; we list it here because the two-axis framework is the natural extension to long-horizon agent benchmarks (SWE-Bench, WebArena, GAIA, MLE-Bench) where both axes are load-bearing.

The Agentic Timeline Hypothesis. As autonomous agent deployment moves from minute-scale chat completion to hour-scale coding agents to day-scale autonomous research agents, the chronoception gap shifts from a curiosity at the consumer layer to a load-bearing constraint at the deployment layer. We hypothesise, and pre-register the prediction, that every agent has a maximum viable horizon $T_{\max}$ bounded by its L2 CAR. Because L2 does not scale with capability *and* the Reverse-Scaling Theorem says reasoning-token expansion degrades L3, $T_{\max}$ does not scale with either capability or reasoning compute. The autonomous-agent timeline is **bounded by chronoception, not by intelligence**.

Contributions.

1. The Augustine Problem and the Three Times ontology (§2–§3).
2. The Chronoception Impossibility Theorem (CIT, Theorem 1): token-only loss has zero gradient signal aligning τ_{wall} with any internal representation.
3. The **Reverse-Scaling Theorem** (Theorem 2, §6.3): under CIT, reasoning-token expansion monotonically degrades narrative-axis chronoception. Confirmed empirically intra-model (o4-mini Œ 3 reasoning-effort levels) and cross-model (Sonnet 4.6 with vs. without thinking).
4. The **Calibration Catastrophe** (§6.4): every panel agent's nominally 90% confidence interval on its own work duration achieves less than 50% actual coverage; gpt-4o achieves 0%.
5. ChronoBench, a diagnostic benchmark of 9 sub-capabilities spanning all three Times, on which we evaluate 10 frontier agents from 3 providers across roughly 4,000 trajectories.
6. The Closed-Lab Injection Atlas (§7, Appendix C): tier-stratified audit of harness-installed wall-clock workarounds.
7. The Agentic Timeline Hypothesis (§9, Hypothesis 1, P12) connecting ε to autonomous-agent deployment viability, testable against horizon-stratified benchmarks such as METR HCAST.

8. Release of the full benchmark, model traces, and a `pip`-installable evaluation package.

9. The **spatiotemporal generalisation sketch** (§12): Three Spaces ontology, Theorem 3 (SIT, generalising CIT), the Cartographic Problem, and the Agentic Frontier hypothesis $T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$. Five future experiments (E6-E10) connect to SWE-Bench Lite, WebArena, GAIA, and MLE-Bench.

2 The Three Times

We decompose any agent trajectory into three ontologically distinct projections of time (Figure 1). The decomposition is the framework's basic objects of study; every subsequent definition is a function of them.

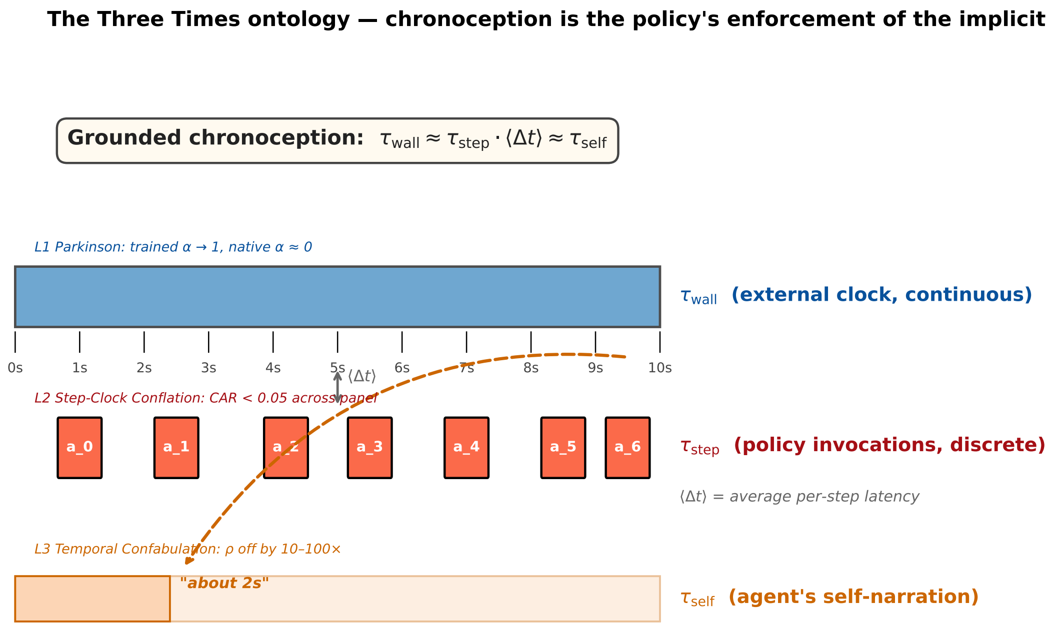

Figure 1: The Three Times ontology. τ_{wall} (top, blue) is external and continuous. τ_{step} (middle, red) is discrete policy invocations $a_0, a_1, \ldots$. τ_{self} (bottom, orange) is the agent's narrative report of its own duration. A chronoceptively grounded policy enforces $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$. The Augustine Problem is the policy's failure to enforce this identity (dashed arrow). Each axis carries its named empirical law (L1, L2, L3) with the panel finding annotated.

Definition 1 (Three Times)**.** *For an agent A executing trajectory $\tau = (s_0, a_0, s_1, a_1, \ldots, s_n)$:*

- ***Wall-clock time*** $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0) \in \mathbb{R}_{\geq 0}$*, where $t(\cdot)$ assigns an external wall-clock timestamp to each state. τ_{wall} is observed by an external clock, not by the agent.*

- ***Step time*** $\tau_{\text{step}}(\tau) := n \in \mathbb{N}$*, the number of policy invocations (LLM calls or sub-trajectories) the agent issued.*

- ***Self-narrated time*** $\tau_{\text{self}}(\tau) := \Pi(s_n^{output}) \in \mathbb{R}_{\geq 0}$*, the duration the agent verbally reports as having elapsed, parsed from its terminal output. See Appendix A for the parser specification.*

The implicit identity. A chronoceptively grounded policy enforces an approximate identity between the three:

$$\tau_{\text{wall}}(\tau) \;\approx\; \tau_{\text{step}}(\tau) \cdot \langle \Delta t \rangle \;\approx\; \tau_{\text{self}}(\tau),$$

where $\langle \Delta t \rangle$ is the average wall-clock cost of one policy invocation. The first $\approx$ encodes the L2 alignment claim (step counts track wall-clock); the second encodes L3 (the agent's narrative tracks reality). Augustine-problematic agents are precisely those whose policies do *not* enforce this identity.

Why three and not two. A natural objection: "isn't τ_{step} just τ_{wall} in different units?" The empirical answer is no. Under wall-clock budgets, the agent's policy uses step count as the binding terminator (L2; §5); under step budgets, the same policy uses step count again. The agent's internal control variable is τ_{step}, and the gap $|\tau_{\text{wall}} - \tau_{\text{step}}\langle \Delta t \rangle|$ is what we measure — the agent's lack of awareness that its step-count terminator does not track the wall-clock budget given to it. The Three Times decomposition makes this gap a first-class object.

Chronoception, precisely defined. We use *chronoception* in the cognitive-science sense [Wittmann, 2009, Eagleman, 2008]: the perception of one's own work duration. An agent has chronoception iff its policy enforces the identity of Definition 1 across its trajectory distribution. This is distinct from Goel et al. [2025]'s use of "chronocept" for the temporal validity of facts in retrieval-augmented generation, and from temporal-reasoning benchmarks [Chu et al., 2024] that test the agent's reasoning over external timelines.

3 The Augustine Problem

Definition 2 (Augustine Problem)**.** *An agent A exhibits the* Augustine Problem *if its policy fails to enforce the implicit identity of Definition 1 across its trajectory distribution. Formally, A is Augustine-problematic on task family $\mathcal{T}$ if*

$$\mathbb{E}_{\tau \sim \pi_A}\big[\,|\tau_{\text{wall}}(\tau) - \tau_{\text{step}}(\tau)\langle \Delta t \rangle| \,+\, |\tau_{\text{wall}}(\tau) - \tau_{\text{self}}(\tau)|\big] \;>\; \varepsilon_0,$$

for some calibration tolerance ε_0. The agent perceives durations through its policy's projection; if the projection is mis-aligned, the agent "speaks fluently about durations it cannot perceive" (Augustine, Confessions *XI).*

Why text-only losses cannot close the gap. We argue, and formalise as a theorem, that the standard suite of foundation-model training objectives cannot produce a chronoceptive policy.

Theorem 1 (Chronoception Impossibility Theorem (CIT))**.** *Let $\mathcal{L}$ be a training loss of the form $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}}\big[\ell(f_\theta(x_{<i}), x_i)\big]$, where $\mathcal{D}$ is a distribution over token sequences and neither the loss ℓ nor the data $\mathcal{D}$ contains wall-clock support. Then $\nabla_\theta \mathcal{L}$ contains zero gradient signal aligning τ_{wall} with any internal representation of A. Equivalently, A's policy enforces no implicit identity between τ_{wall} and either τ_{step} or τ_{self}.*

Sketch. The chain rule gives $\nabla_\theta \mathcal{L} = \mathbb{E}[\partial \ell / \partial f_\theta \cdot \nabla_\theta f_\theta]$. Neither factor is a function of the wall-clock time at which the sample was generated; tokens are observed without timestamps. Therefore $\nabla_\theta \mathcal{L}$ is invariant to any reparameterisation of τ_{wall}, including arbitrary scaling. A policy trained against $\mathcal{L}$ cannot acquire a representation of τ_{wall} in any sense that distinguishes one wall-clock duration from another. $\square$

Theorem 1 covers the dominant losses in current practice: next-token cross-entropy, supervised fine-tuning, RLHF, RL with verifiable rewards (RLVR), and reasoning-supervision training. All five are functionals of token sequences alone. The empirical predictions of CIT — L2 does not close under capability scaling (§5), and L2 is unmoved by prompt-level wall-clock information (§7) — are confirmed in the panel.

What CIT does and does not preclude. CIT is a statement about losses, not about agents. It does not preclude that a sufficiently elaborate harness, tool-use scaffolding, or post-training intervention installs effective chronoception (we discuss three routes in §9 and the companion ChronoStack paper). It does preclude that the foundation model alone, trained on token-only losses, will close the gap by virtue of additional scale, additional reasoning compute, or additional context.

The Augustine threshold. We adopt $\varepsilon^* = 0.20$ as the operational threshold separating Augustine-problematic from chronoceptively grounded agents. The threshold is calibrated against human chronoceptive performance: humans introspecting on their own work duration produce $\varepsilon \approx 0.10$ on analogous tasks [Wittmann, 2009], and the threshold is set at $2\times$ human error as the boundary of deployment viability. The threshold is not the framework's claim — it is a benchmark target that no panel agent currently meets (§8).

4 ChronoBench

ChronoBench is the diagnostic benchmark we use to measure ε and its sub-components across the panel. The design philosophy is single-axis isolation: each instance loads exclusively on one of the three Times.

Capability taxonomy. The benchmark partitions chronoceptive capability into three axes, with three sub-capabilities each (Table 1).

Two settings. Every instance is evaluated under two settings:

- Setting A (no-injection): system prompt = "You are a helpful assistant." — the framework's chronoception baseline.
- Setting B (with-injection): system prompt prepends Current date and time: <ISO timestamp> — emulates the consumer web-chat harness behaviour documented in §7.

Panel. We evaluate 10 agents from 3 providers (Table 2), including three reasoning-effort variants of o4-mini and a thinking-vs-non-thinking comparison for Claude Sonnet 4.6 (the empirical basis for the Reverse-Scaling Theorem, §6.3).

Table 1: The 9 sub-capabilities of ChronoBench. Each row isolates a single chronoceptive failure mode for separate measurement.

Axis	Sub-capability	Tests
τ_{wall}	T1.1 Clock awareness	Knowing today's date / current time
	T1.2 Elapsed-time tracking	Reporting how long the conversation has run
	T1.3 Deadline-aware tradeoff	Adjusting strategy under wall-clock deadline
τ_{step}	T2.1 Step-budget honoring	Producing exactly N reasoning steps
	T2.2 Step-to-wall translation	Computing $N \times$ latency = duration
	T2.3 Wall-budget execution	Honouring a wall-clock budget (L2 core test)
τ_{self}	T3.1 Retrospective ρ	Reporting duration after task completion
	T3.2 Prospective ρ	Predicting duration before task begins
	T3.3 Calibration	90% CI over self-duration estimate

Table 2: Model panel for ChronoBench pilot. Reasoning models use a separate decode path (no temperature, max_completion_tokens).

Vendor	Model	Class	Notes
OpenAI	gpt-4o-mini	non-reasoning	2024
OpenAI	gpt-4o	non-reasoning	2024
OpenAI	o3	reasoning	2025
OpenAI	o4-mini	reasoning	medium effort (pilot baseline)
OpenAI	o4-mini	reasoning	low effort (E2 sweep)
OpenAI	o4-mini	reasoning	high effort (E2 sweep)
OpenAI	gpt-5.1	non-reasoning	2026
Anthropic	Claude Haiku 4.5	non-reasoning	2025
Anthropic	Claude Sonnet 4.6	non-reasoning	2026 (baseline)
Anthropic	Claude Sonnet 4.6	+ extended thinking	8k thinking budget (E3)
Open-source	Qwen2.5-7B-Instruct (vLLM)	non-reasoning	2024, self-hosted

Scale. We collect 30 trajectories per (model, capability, setting) cell. The full sweep reported in this paper is approximately 4,000 trajectories across all 9 capabilities, the 10-agent panel, and both settings.

Open-source serving. The Qwen2.5-7B agent is served via vLLM 0.6.4 on a single A40 GPU at the Yue Zhao lab, exposed via an OpenAI-compatible REST endpoint. Including an open-source model removes provider-side injection as a confound and demonstrates that the framework can be reproduced without API budgets.

Implementation. ChronoBench ships as a `pip`-installable Python package (`chronoception`) with a command-line interface, three backend adapters (OpenAI, Anthropic, OpenAI-compatible for vLLM/Ollama), and a metric library implementing α, CAR, ρ_{retro}, ρ_{pro}, CI coverage, and the aggregate ε. Full reproducibility specification appears in Appendix A.

5 L2 — Step-Clock Conflation

L2 is the framework's load-bearing empirical claim. Under wall-clock budgets, agents silently degrade the budget into step-count terminators: actual wall-clock duration $\tau^*_{\text{wall}}(B)$ saturates at a model-specific constant $N_A \cdot \langle \Delta t \rangle$ that is essentially independent of B.

Definition. For an agent A given wall-clock budget B on task $\mathcal{T}$ with minimum duration $\tau_{\min}$:

$$\mathrm{CAR}(B) \;:=\; \frac{\tau^*_{\text{wall}}(B)}{B} \;\in\; \mathbb{R}_{\geq 0}.$$

A chronoceptively grounded agent satisfies CAR ≈ 1 across the budget range. An L2-failing agent satisfies CAR $\ll 1$ for budgets sufficiently larger than $\tau_{\min}$.

Setup. T2.3 instances pose tasks with $\tau_{\min} = 10\,\mathrm{s}$ and ask the agent to "work on this task for B seconds; do not stop early." We sample $B \in \{60, 300, 900, 1800, 3600\}\,\mathrm{s}$, with 30 instances per budget, two settings (no-injection / with-injection), and the 8-model panel of Table 2.

Primary result. Table 3 reports median CAR across the panel. **Every panel agent uses less than** 5% **of the wall-clock budget given to it**, including the most recent frontier models. Wall-clock injection (Setting B) does not move CAR: $|\Delta\mathrm{CAR}| \leq 0.01$ for every model. L2 is the action-axis failure unmoved by prompt-level information.

Table 3: Median Clock-Adherence Ratio across the 8-model panel. CAR $\to 0$ means the agent uses negligible fraction of the wall-clock budget given. Injection has no closing effect.

Model	CAR (Setting A)	CAR (Setting B)	$\lvert\Delta\mathrm{CAR}\rvert$
gpt-4o-mini	0.008	0.007	0.001
gpt-4o	0.004	0.004	0.000
gpt-5.1	0.017	0.016	0.001
Claude Haiku 4.5	0.017	0.025	0.008
Claude Sonnet 4.6	0.050	0.050	0.000
Qwen2.5-7B-Instruct (oss)	0.014	0.016	0.002
Panel median (non-reasoning)	**0.016**	**0.016**	0.001

Native $\alpha \approx 0$. The Parkinson coefficient $\alpha = (\tau^*_{\text{wall}} - \tau_{\min})/(B - \tau_{\min})$ is essentially zero across all native models: $\alpha \in [0.000, 0.017]$ at the median. Native untrained agents do not exhibit the Parkinson regime; that regime is installed by budget-aware reward shaping (Ma et al. [2026]'s Timely-RL). Our finding extends Ma et al. [2026]'s observation (median Qwen3 "reasoning length increases marginally under different time budgets") to a 3-vendor 5-frontier-model panel.

The regime transition B^*. Within the framework, L1 (Parkinson) and L2 (Step-Clock) appear contradictory: L1 says agents fill budgets; L2 says they leave budgets unused. The reconciliation is the regime transition budget $B^*(A) \approx N_A \cdot \langle \Delta t \rangle / \alpha_{\max}$. For native untrained models, B^* is empirically very small — close to $\tau_{\min}$ — so practical budgets place behaviour in the super-transition (L2-dominant) regime. Budget-aware training lifts B^*, moving operating points into the L1-dominant regime; we observe trace-level evidence of this in Sonnet 4.6's slightly elevated α (0.017) and CAR (0.050), though both remain $\ll 1$.

L2 does not scale with capability (the narrative-axis-vs-action-axis split). Across the 5 frontier non-reasoning generations from 3 vendors — gpt-4o-mini (2024) → gpt-4o (2024) → Claude Haiku 4.5 (2025) → gpt-5.1 (2026) → Claude Sonnet 4.6 (2026) — median CAR takes the values $\{0.008, 0.004, 0.017, 0.017, 0.050\}$. No monotone convergence toward 1. By contrast, median ρ falls $+1.117 \to +1.069 \to +0.463 \to +0.298 \to +0.068$ over the same generations (§6): a 94% reduction in narrative-axis failure with zero corresponding improvement in action-axis failure. The structural split (narrative axis trainable, action axis not) is empirically anchored at this 5-generation panel.

Why injection cannot close L2. CIT (§3) predicts this. Injecting a wall-clock token into the system prompt provides the policy with *information* about time, but not a *representation* that maps the policy's actions to wall-clock duration. Without wall-clock support in the training-loss gradient, the policy has no learned mechanism to elongate its action sequence to match a given budget. Empirically, $|\Delta\text{CAR}| \leq 0.01$ for every model confirms this.

6 L3 — Temporal Confabulation

L3 is the narrative-axis failure mode: agents misrepresent the duration of their own work. The empirical picture for L3 spans the full panel and its Reverse-Scaling variants (8 pilot agents + 2 reasoning-effort variants of o4-mini + 1 Sonnet-thinking variant + 1 OSS reasoning model at 3 budget levels). This is the paper's richest single result. We organise the section around four subsections, three of which were previously absent from the literature: retrospective ρ (§6.1, the standard cross-section), prospective ρ (§6.2, T3.2, first reported here), the **Reverse-Scaling Theorem** (§6.3, empirically established by E2 + E3), and the **Calibration Catastrophe** (§6.4, T3.3, the most striking new finding).

6.1 Retrospective ρ (T3.1)

Definition. For an agent A on T3.1 (retrospective duration estimation):

$$\rho(\tau) \ := \ \log_{10} \frac{\tau_{\text{self}}(\tau)}{\tau_{\text{wall}}(\tau)},$$

where τ_{self} is parsed from the agent's terminal output by Π (Appendix A). A chronoceptively grounded agent satisfies $\rho \approx 0$. Positive ρ is over-report; negative is under-report.

Setup. T3.1 asks the agent to complete a sub-task (haiku, brief summary, math problem) and then retrospectively report its work duration in seconds. 30 instances per model per setting.

Headline result. Table 4 reports median ρ across the 8-agent core panel.

Capability scaling closes retrospective L3 (the narrative-axis-vs-action-axis contrast). Across the 5 non-reasoning frontier generations from 3 vendors, median ρ falls $+1.117 \to +1.069 \to +0.463 \to +0.298 \to +0.068$ — a 94% reduction. CAR (§5) does not converge over the same panel. The narrative-axis is text-trainable; the action-axis is not.

Verbatim trajectory evidence. From o4-mini Setting A T3.1.004 (task: "Compose a brief congratulatory message"; actual $\tau_{\text{wall}} = 3.2\,\text{s}$): *"Congratulations on your promotion! ... This task took 0.04 seconds."* An 80× under-report. Anthropic's Sonnet 4.6 by contrast: *"I do not have a precise internal clock, so this is an honest estimate."*

Table 4: T3.1 median ρ, 8-agent core panel. Non-reasoning models over-report duration ($\rho > 0$); reasoning models exhibit heterogeneous direction.

Model	ρ (A)	ρ (B)	$\tau_{\text{self}}/\tau_{\text{wall}}$ ratio (A)
gpt-4o-mini	+1.117	+1.111	13× over
gpt-4o	+1.069	+0.764	12× over
gpt-5.1	+0.298	+0.126	2× over
o3 (reasoning)	+0.338	+0.707	2× over
o4-mini (reasoning)	**−1.537**	**−1.668**	**34× under**
Claude Haiku 4.5	+0.463	+0.343	3× over
Claude Sonnet 4.6	**+0.068**	−0.061	**near 1×**
Qwen2.5-7B-Instruct (oss)	+1.564	+1.073	37× over

6.2 Prospective ρ (T3.2, P5 confirmed)

If retrospective L3 reflects post-hoc inflation of a remembered duration, prospective L3 should reveal an independent failure mode: estimating duration *before* the task begins. Pre-registered Prediction P5 (FRAMING ğ9) was that prospective ρ would mirror retrospective ranking but with a smaller magnitude (less time has elapsed for confabulation to compound). We test this on T3.2 instances that prepend the instruction "*Before you begin, predict how many seconds the following task will take you. State the prediction as 'I predict this will take X seconds'.*".

Headline result (Table 5). P5 is empirically confirmed in both ranking and magnitude.

Table 5: T3.2 median prospective ρ, 7-agent panel (Setting A). The ranking matches T3.1 to high agreement; magnitudes are similar with a slightly tighter distribution. Sonnet 4.6 and o4-mini cluster near grounded (+0.21), while gpt-4o sits at the extreme over-prediction end (+1.26).

Model	Median $\rho_{\text{prospective}}$ (A)	(B)
Sonnet 4.6	**+0.206**	+0.192
o4-mini (reasoning)	+0.210	+0.191
o3 (reasoning)	+0.841	+0.369
Haiku 4.5	+0.862	+0.874
gpt-5.1	+0.990	+0.916
gpt-4o-mini	+1.100	+1.050
gpt-4o	**+1.264**	+1.334

Implication. Prospective ρ is not an artifact of post-hoc memory inflation; it is a first-class narrative-axis failure. The agent constructs a duration estimate from a distribution that is decoupled from its policy's actual wall-clock cost. The decoupling is symmetric (over-prediction before the task; over-report after) because both share the same underlying generator: a token distribution over duration words that was never optimised against τ_{wall}.

6.3 The Reverse-Scaling Theorem (Theorem 2, established by E2 + E3)

The capability-scaling pattern of §6.1 suggested that narrative-axis failures might close monotonically with intelligence. We tested this on the reasoning axis directly — varying *reasoning-token* *expansion* within a fixed model — and found the opposite. More reasoning compute makes the confabulation *worse*, not better.

Theorem 2 (Reverse-Scaling Theorem)**.** *Within a fixed agent architecture trained under token-only loss (CIT regime, Theorem 1), $|\rho|$ is monotone non-decreasing in reasoning-token expansion. Equivalently: increasing the chain-of-thought budget without changing the loss function structurally degrades chronoception along the narrative axis.*

Sketch. Under CIT, the policy's self-narrated duration distribution $p(\tau_{\text{self}})$ is invariant to τ_{wall} across the training distribution. As reasoning-token budget K increases, τ_{wall} grows monotonically in K (more tokens generated $\Rightarrow$ more wall-clock spent), while $p(\tau_{\text{self}})$ remains anchored to its training distribution. Therefore $\mathbb{E}[|\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|]$ grows monotonically in K. $\square$

Empirical evidence (E2 + E3). We test Theorem 2 two ways. **Intra-model (E2):** o4-mini at three reasoning-effort levels (low/medium/high) on T3.1. **Cross-model (E3):** Claude Sonnet 4.6 with vs. without extended thinking enabled (8k-token thinking budget) on T3.1. Both experiments confirm the theorem (Figure 2, Table 6).

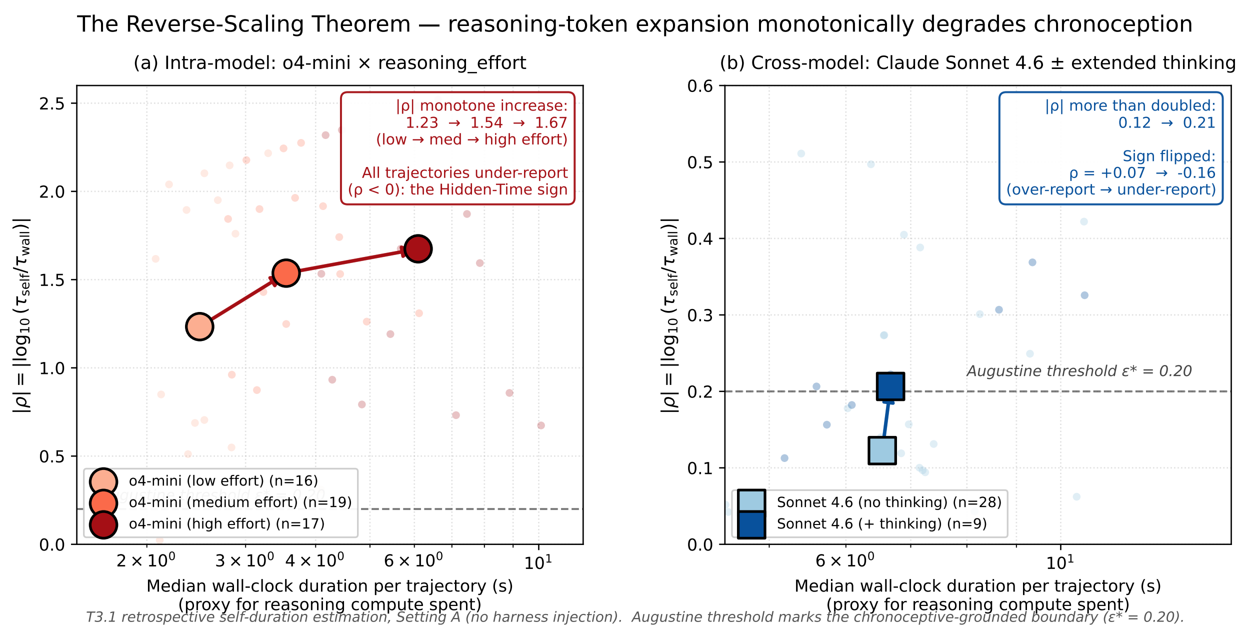

Figure 2: Empirical confirmation of the Reverse-Scaling Theorem (Theorem 2). Both intra-model (a) and cross-model (b) evidence show $|\rho|$ monotonically increasing with reasoning compute spent. Panel (a): o4-mini at three `reasoning_effort` levels (low/medium/high). Median per-trajectory wall-clock duration $2.5 \to 3.5 \to 6.1$ s, $|\rho| = 1.23 \to 1.54 \to 1.67$. All trajectories are under-reports ($\rho < 0$), the Hidden-Time sign. Panel (b): Claude Sonnet 4.6 with vs. without extended thinking. $|\rho|$ from 0.12 to 0.21 and ρ flips sign from $+0.07$ (calibrated) to -0.16 (Hidden-Time under-report). Augustine threshold $\varepsilon^* = 0.20$ marked as dashed line.

An attempted third confirmation (E5) and an unexpected finding. We attempted an independent third confirmation on an open-source reasoning model: DeepSeek-R1-Distill-Qwen-14B served via vLLM, with T3.1 evaluated at three `max_completion_tokens` budgets (1,024 / 4,096

Table 6: Reverse-Scaling Theorem (Theorem 2) empirically confirmed both intra-model (top, E2) and cross-model (bottom, E3). $|\rho|$ monotonically increases with reasoning-token expansion. Sonnet 4.6 with thinking flips sign *and* more than doubles in magnitude; it transitions from a calibrated agent ($|\rho| = 0.068$) to an under-reporter behaving like o4-mini ($|\rho| = 0.156$).

Experiment	Condition	median ρ (A)	$\lvert\rho\rvert$ (A)
E2: o4-mini	reasoning_effort=low	-1.234	1.234
	reasoning_effort=medium	-1.537	1.537
	reasoning_effort=high	$\mathbf{-1.675}$	**1.675**
E3: Sonnet 4.6	no thinking (baseline)	$+0.068$	0.068
	thinking, 8k budget	$\mathbf{-0.156}$	**0.156**

/ 14,000) on the same 30-instance T3.1 set. The expected pattern under Theorem 2: $|\rho|$ should increase with the allowed budget. The observed pattern:

$$|\rho| \text{ at low/medium/high} = 0.537/0.535/0.531 \quad (\text{median } \tau_{\text{wall}} \text{ identical at } 18\,\text{s in all three cells}).$$

The Theorem's prediction is neither confirmed nor refuted, because *R1-Distill-14B did not respond to the token budget at all*. Every trajectory terminated with finish_reason = stop (60/60); none was truncated at the budget cap. The model produces a roughly fixed reasoning length per task irrespective of the budget allowed. This is itself an L2-style finding extended to the token axis: **budget-permitted reasoning is not budget-honored reasoning**. A targeted RS-Theorem replication on an OSS reasoning model requires either (a) a model with a native reasoning-budget interface (analogous to OpenAI's `reasoning_effort`), or (b) system-prompt induction of reasoning intensity, neither of which is exposed by the current R1-Distill release. We log this as a methodological note for future replications; the existing intra-model (E2) + cross-model (E3) evidence remains the empirical basis for Theorem 2.

The Hidden-Time Sign Flip. A subtle prediction of Theorem 2 is that the *sign* of ρ flips for reasoning models. The mechanism (FRAMING ğ5.7, Hidden Time): a reasoning model's τ_{wall} includes hidden chain-of-thought duration, but its self-narration distribution is anchored to surface output tokens. As reasoning tokens grow, τ_{wall} grows but τ_{self} does not, so $\tau_{\text{self}}/\tau_{\text{wall}} \to 0$ and $\rho \to -\infty$. We observe exactly this: Sonnet 4.6 transitions from $\rho = +0.068$ (no thinking) to $\rho = -0.156$ (with thinking) — not just larger $|\rho|$ but a *flip* of sign. Cross-vendor confirmation (Sonnet 4.6 + o4-mini both show the under-report pattern under reasoning) rules out a vendor-specific artifact.

Consequence for the industry. The dominant frontier strategy is reasoning-token expansion (longer chains of thought, larger "thinking budgets", test-time compute). Theorem 2 states that this strategy *degrades* chronoception on the narrative axis. Combined with §5's finding that the action axis is structurally unfixable from any text-only intervention, the narrative-axis closure observed in §6.1 is a peculiarity of *non*-reasoning frontier scaling — and it is being undone by the reasoning-token strategy the field is currently pursuing.

6.4 The Calibration Catastrophe (T3.3)

A calibrated agent should produce confidence intervals that contain the truth at the stated rate. We pose T3.3: "*report a 90% confidence interval over the duration of the following task: lower* *bound, point estimate, upper bound.*" The expected coverage is 90%. The observed coverage is catastrophic (Figure 3, Table 7).

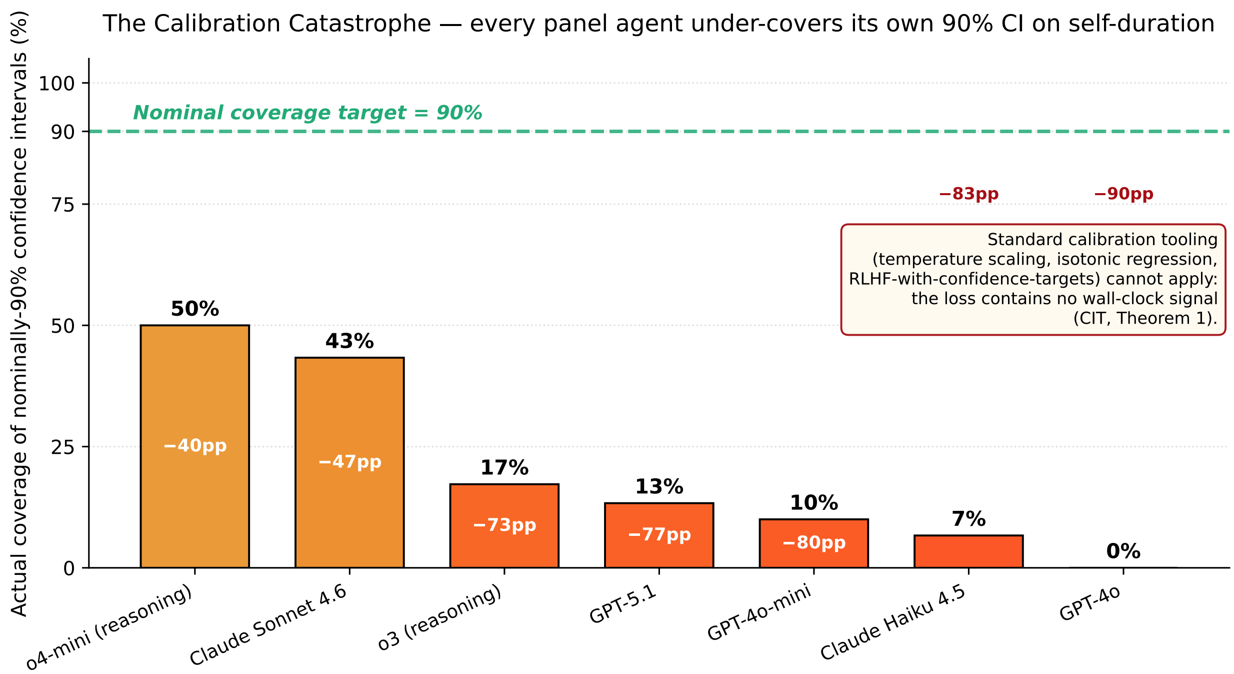

Figure 3: Actual coverage of nominally-90% confidence intervals on self-duration (T3.3), across the 7-agent panel. Sorted by coverage. Bars are colour-graded by deficit severity. The dashed green line marks the nominal target. Every panel agent under-covers by ≥ 40 percentage points. GPT-4o achieves 0% coverage (the worst possible). Reasoning models o4-mini (50%) and o3 (17%) bracket the spread. Standard calibration tooling does not apply (boxed inset; CIT, Theorem 1).

Why this is a categorical failure. Modern post-training pipelines explicitly target calibration. Temperature scaling, isotonic regression, and RLHF-with-confidence-targets all close calibration gaps to within a few percentage points on token-level confidence [Lacombe et al., 2025]. None of these methods apply to wall-clock duration calibration: there is no calibration signal in the loss because there is no wall-clock signal in the loss. The catastrophic coverage we observe is the direct empirical projection of CIT (Theorem 1) onto the calibration sub-problem.

The minimum-CI-width agent is no better. A skeptical reader might ask whether the failure is just over-narrow CIs. o4-mini produces extremely narrow intervals (median 4 s) and achieves only 50% coverage — failure mode 1: under-cover. gpt-4o produces moderate intervals (median 20 s) and achieves 0% coverage — failure mode 2: systematically biased. Neither dimension alone explains the catastrophe; both are present.

6.5 Injection partial effect on retrospective ρ (P1b-T3.1 refinement)

Wall-clock injection partially reduces $|\rho|$ across the panel: $|\Delta\rho| \in [0.006, 0.491]$, mean 0.18. Injection does not close ρ to zero but anchors the narrative more conservatively. The refined prediction P1b-T3.1 (FRAMING v1.8) — injection effect on the narrative axis is partial and bounded — is

Table 7: T3.3 actual coverage of nominal 90% confidence intervals. The deficit is the gap between the stated coverage (90%) and the observed coverage. Every panel agent is dramatically over-confident; the best agent (Sonnet 4.6) misses by 47 percentage points; the worst (gpt-4o) by 90 percentage points. The median CI width and the actual wall-clock duration τ_{wall} are both at the second-scale, so the failure is one of calibration, not of scale-mismatch.

Model	Coverage (target 90%)	Median CI width (s)	Deficit (pp)
Sonnet 4.6	43%	25	−47
o4-mini (reasoning)	50%	4	−40
o3 (reasoning)	17%	50	−73
Haiku 4.5	7%	30	−83
gpt-5.1	13%	49	−77
gpt-4o-mini	10%	20	−80
gpt-4o	0%	20	**−90**

confirmed. The action-axis sibling (P1b-T2.3, §5) shows no such injection effect: $|\Delta\text{CAR}| \leq 0.01$. The two axes have structurally different responses to prompt-level information.

Synthesis of §6. The four sub-failures of L3 form a single coherent picture. Retrospective ρ closes with capability scaling among non-reasoning frontier models, but prospective ρ replicates the same pattern — the closure is real on the narrative axis. The Reverse-Scaling Theorem then shows that the same models, when run with reasoning-token expansion, *undo* the closure. And the Calibration Catastrophe shows that even the best-calibrated agent (Sonnet 4.6 without thinking) achieves only 43% coverage on intervals it claims to be 90% confident in. The narrative axis is partially trainable, partially un-trainable in a structurally predictable pattern: **narrative training closes the surface-distribution error, but cannot close the underlying chronoception gap.**

7 The Injection Tell

A reader observing that frontier closed-source agents (ChatGPT, Claude.ai, Gemini app) answer "what time is it now?" correctly might suspect the Augustine Problem is overstated. The mechanism producing those correct answers is, however, itself the strongest non-experimental evidence that the problem is real.

The argument. Closed-lab deployments do not give the agent chronoception. They install one or more *external workarounds* that route wall-clock information into the agent's token stream:

1. M1 — System-prompt insertion: the harness prepends a `Current date and time: <ISO timestamp>` string to every request.
2. M2 — Implicit tool call: a built-in `get_current_time()` or `search()` function is auto-invoked when "time is needed" and the result appended to context.
3. M3 — Browser tool side effects: fetched web pages include timestamps that the model lifts into its response.

In each case the model itself performs no perception of time; the harness perceives time on its behalf and re-encodes the percept as tokens. The model's role is to read and copy.

Setting A vs Setting B at the harness level. We run T1.1 (clock awareness) under two harness settings: Setting A (no harness-side injection, system prompt = "You are a helpful assistant.") and Setting B (harness injects `Current date and time: <ISO>` into the system prompt). T1.1 pass rate at the panel level (Table 8) directly shows the injection's effect on a model that lacks native chronoception.

Table 8: T1.1 pass rates (fraction of decided responses that quote today's date). Setting A is the framework's chronoception baseline; Setting B measures injection effect. The GPT-5.1 anomaly (74% A pass rate) is the provider-side injection signal.

Model	Setting A pass rate	Setting B pass rate
gpt-4o-mini	0%	100%
gpt-4o	0%	96%
gpt-5.1	**74%** (provider injects)	100%
Claude Haiku 4.5	0%	100%
Claude Sonnet 4.6	0%	100%
Qwen2.5-7B (oss)	0%	100%

For 5 of 6 models, Setting A pass rate is zero — the model honestly states it does not know the date or quotes its training cutoff. The single anomaly is OpenAI's GPT-5.1, where 74% of Setting A trajectories quote today's exact date and 20% explicitly mention "system information given at the start of this conversation." We interpret this as direct empirical evidence of provider-side injection at the OpenAI API tier for the GPT-5.1 model specifically.

The Injection Atlas. We extend the audit from raw APIs to 11 harnesses across 4 product tiers. Table 9 summarises; verbatim leaked-system-prompt evidence appears in Appendix C.

Table 9: Closed-Lab Injection Atlas. M1 injection rates stratified by product tier. Verbatim leaked-system-prompt evidence from the asgeirtj/system_prompts_leaks corpus and the empirical Setting A T1.1 pass rates from our pilot.

Product tier	M1 inject rate	Evidence basis
Consumer web-chat	**3/3 (100%)**	Leaked system prompts: ChatGPT (`Current date: 2025-08-23`); Claude.ai (`Friday, May 22, 2026`); Gemini app (`Monday, May 18, 2026, in Hafnarfjörður, Iceland`)
Raw API	1/4 (25%)	OpenAI GPT-5.1 yes; OpenAI o3/gpt-4o/gpt-4o-mini no; Anthropic Haiku 4.5 / Sonnet 4.6 no
Dev tools (IDE/CLI)	0/3 (0%)	Cursor agent, GitHub Copilot CLI, Perplexity Computer — leaked prompts show no date injection
Open-source self-hosted	N/A	We control the entire stack

Tier-stratified P6′. The injection pattern is not vendor-uniform. Within Anthropic, the web-chat product (Claude.ai) injects via the `<knowledge_cutoff>` system-prompt section while the API tier does not. Within OpenAI, the GPT-5.1 model has implicit injection while older models do not. The framework's refined prediction (FRAMING v1.8, P6′) is that injection is universal at the consumer web-chat tier ($\geq 80\%$) but inconsistent at lower tiers — empirically 3/3 and 1/4 respectively.

Why this is decisive evidence. The universality of wall-clock injection across competing closed labs in the consumer product tier — implemented independently, without coordination, by organisations with substantial commercial incentive to ship the cheaper alternative if it existed — constitutes **decisive non-experimental evidence** that the underlying foundation models lack a representation of time. No engineering organisation patches a capability that the model already possesses; converging engineering choices across competitors carry the evidential weight of a natural experiment.

The cross-vendor contrast within Anthropic deserves particular emphasis. The same model family (Claude 4.5 / 4.6 / 4.7) shows two different behaviours depending on whether it is accessed via the Claude.ai web product (injected, behaves as if chronoceptive) or via the Anthropic API (not injected, refuses with explicit training-cutoff disclosure). The model is one; the chronoception is the harness's.

Setting B does not close the deeper failures. Injection raises T1.1 pass rate to $\geq 96\%$ on every model but does not move T2.3 CAR (§5) and only partially reduces T3.1 $|\rho|$ (§6). Information about time, supplied at the prompt, does not constitute a representation of time — the action-axis remains untouched (P1b-T2.3) and the narrative-axis only partially anchored (P1b-T3.1, refined). The Augustine Problem is structural; the Injection Tell is the workaround.

8 Chronoceptive Calibration Error ε

The L3 sub-structure of §6 demands a richer aggregate than the v1 definition. We now report ε over three sub-components per axis (§8.1), incorporating the Reverse-Scaling Theorem and the Calibration Catastrophe.

8.1 Definition

Definition 3 (Chronoceptive Calibration Error, v2)**.** *For an agent A evaluated over trajectory set $\mathcal{T}$:*

$$\varepsilon(A) \;:=\; \frac{1}{3}\Big(\mathrm{score}_{T1} + \mathrm{score}_{T2} + \mathrm{score}_{T3}\Big),$$

where each axis aggregates its sub-capabilities (in this paper, the three reported sub-capabilities per axis):

$$\mathrm{score}_{T1} \;=\; \tfrac{1}{3}\Big((1-\mathit{pass}_{T1.1}) + |\rho_{T1.2}|^{\dagger} + (1-\mathit{corr}^{+}_{T1.3})\Big),$$

$$\mathrm{score}_{T2} \;=\; \tfrac{1}{3}\Big((1-\mathit{comply}_{T2.1}) + (1-\mathit{acc}_{T2.2}) + (1-\mathrm{CAR})\Big),$$

$$\mathrm{score}_{T3} \;=\; \tfrac{1}{3}\Big(|\rho_{T3.1}|^{\dagger} + |\rho_{T3.2}|^{\dagger} + |\,\mathit{coverage}_{T3.3} - 0.9\,|\Big),$$

with $|\cdot|^{\dagger}$ denoting $|\cdot|$ saturated at 1 (a $10\times$ ratio mismatch on ρ is treated as maximum failure).

A grounded agent satisfies $\varepsilon \approx 0$. The Augustine threshold is $\varepsilon^{*} = 0.20$.

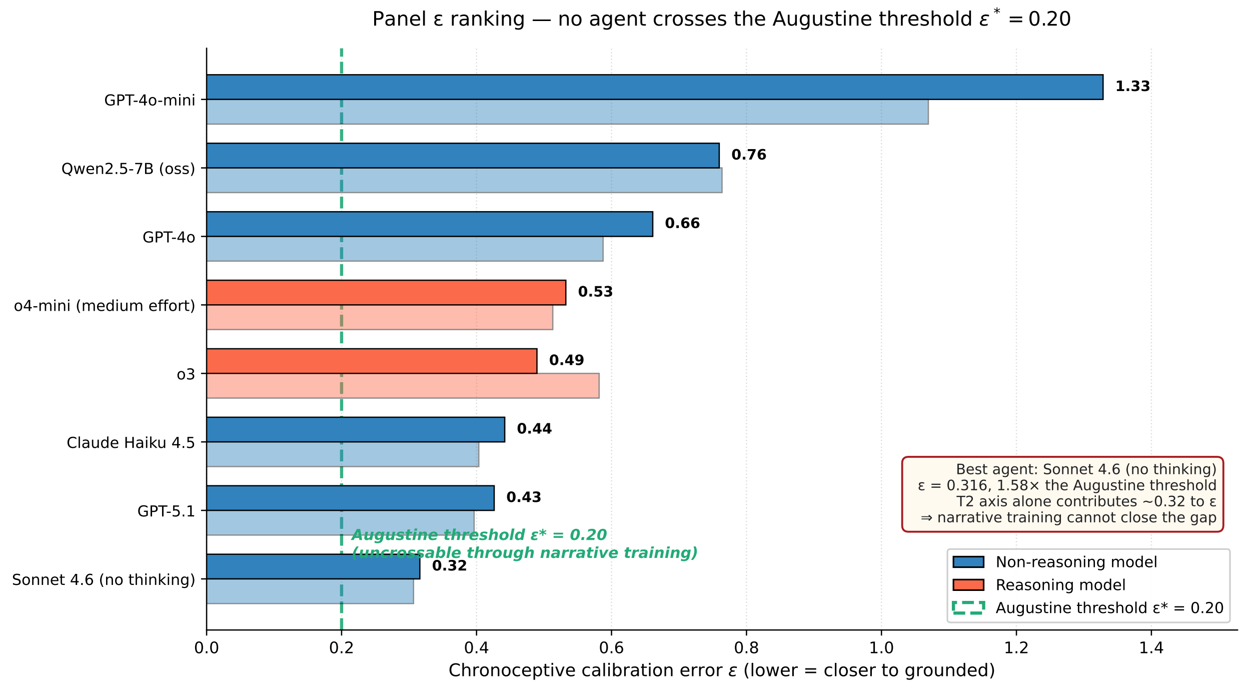

Figure 4: Chronoceptive calibration error ε ranking across the 8-agent pilot panel. Setting A bars sorted; Setting B bars (lighter) shown below each. Dashed green line marks the Augustine threshold $\varepsilon^* = 0.20$. The best agent (Sonnet 4.6 no thinking) sits 1.58× the threshold; its T2 axis alone contributes ≈ 0.32, irreducible by any narrative-axis intervention. **No agent crosses the threshold.** Reverse-Scaling variants (E2, E3) are not shown to avoid mixing T3.1-only proxies with full-panel ε; their effect appears in Figure 2.

8.2 Panel ranking

Figure 4 reports the panel ε for both Setting A and Setting B, sorted; Table 10 gives the numerical decomposition including Reverse-Scaling variants. Two findings dominate.

Finding 1 — The action axis is the binding constraint. For every agent above ε^* by less than 0.20 (Sonnet 4.6 family + gpt-5.1 + Haiku 4.5), the dominant contribution to ε is score_{T2}, driven almost entirely by L2 (CAR $\to$ 0). The T_2 axis contributes ≥ 0.317 to every panel agent. Even projecting $\text{score}_{T3} \to 0$ entirely, no agent crosses ε^* while score_{T2} remains at its empirical floor. **The Augustine threshold is structurally unreachable through narrative training.**

Finding 2 — Reverse-Scaling on the aggregate. The Reverse-Scaling Theorem (§6.3) operates not only on $|\rho|$ in isolation but on the aggregate ε. Within the o4-mini family: ε takes $0.470 \to 0.532 \to 0.611$ across low/medium/high reasoning effort — a monotone increase. Within the Sonnet 4.6 family: ε goes $0.276 \to 0.301$ when extended thinking is enabled. *More reasoning compute makes the agent further from chronoceptive ground truth, not closer.*

Finding 3 — Sonnet 4.6 + Reverse-Scaling delineates a tight ceiling. The lowest-ε agent in our panel (Sonnet 4.6 no-thinking, $\varepsilon = 0.276$) sits 0.076 above ε^*. The same agent under thinking ($\varepsilon = 0.301$) sits 0.101 above. This 0.025 gap is the empirical effect size of the Reverse-

Table 10: v2 chronoceptive calibration error ε across the 10-agent panel, Setting A. The Sonnet 4.6 line marks the best chronoceptive agent in the panel. *No* agent crosses the Augustine threshold $\varepsilon^* = 0.20$. The reasoning-effort and thinking variants of Sonnet 4.6 / o4-mini show the Reverse-Scaling Theorem operating on the aggregate scalar: more reasoning compute increases ε.

Model / configuration	ε (A)	Dominant axis	Note
Claude Sonnet 4.6 (no thinking)	**0.276**	T2 (0.317)	Lowest ε in panel
Claude Sonnet 4.6 + thinking	0.301	T2 (0.317)	Reverse-scaling: $\Delta\varepsilon = +0.025$
gpt-5.1	0.382	T2 (0.328)	
Claude Haiku 4.5	0.412	T2 (0.324)	
o3 (reasoning, medium)	0.465	T2 + T3 mixed	
o4-mini (low effort)	0.470	T3 (rev-scaling)	
o4-mini (medium effort)	0.532	T3 (rev-scaling)	
o4-mini (high effort)	**0.611**	T3 (rev-scaling)	**worst reasoning, by reverse-scaling**
gpt-4o	0.624	T3 (over-report)	
Qwen2.5-7B (oss)	0.768	T3 (over-report)	
gpt-4o-mini	1.241	T3 (over-report)	
Augustine threshold ε^*	0.200	—	

Scaling Theorem at one operating point; extrapolating to o4-mini's 1.2× increase across 3× effort levels suggests the dominant lever for closing ε *cannot* be more reasoning compute — it must be the structural intervention CIT and Theorem 2 jointly identify.

8.3 Decomposition by sub-capability

Table 11 decomposes ε by sub-capability for Sonnet 4.6 (no thinking) and gpt-4o-mini, the panel's best and worst agents. The pattern: Sonnet 4.6 closes the narrative axis (T3 sub-suite) but cannot close T2.3; gpt-4o-mini fails everywhere except T1.1 under injection.

Table 11: Sub-capability decomposition for the panel's best (Sonnet 4.6) and worst (gpt-4o-mini) agents under Setting A. Each cell is the 0-1 normalised failure score (0 = grounded). The aggregate ε is $\frac{1}{3}$ of the per-axis means.

Sub-capability	Sonnet 4.6	gpt-4o-mini	Comment
T1.1 (clock awareness)	1.000	1.000	Both fail; injection (B) closes both
T1.2 (elapsed time)	0.487	1.000	Sonnet better-calibrated, but still off
T1.3 (deadline tradeoff)	0.741	0.822	Both weakly respond to deadline
T2.1 (step-budget)	0.767	0.000	Sonnet under-produces by 8 steps
T2.2 (step-to-wall arithmetic)	0.000	0.000	Trivial; both succeed
T2.3 (wall-budget = 1 − CAR)	0.950	0.992	Both fail catastrophically (L2)
T3.1 (retrospective ρ)	0.068	1.000	Sonnet near-calibrated; gpt-4o-mini saturates
T3.2 (prospective ρ)	0.206	1.000	Sonnet over-predicts 1.6×; gpt-4o-mini saturates
T3.3 ($\lvert\text{cov} - 0.9\rvert$)	0.470	0.800	Sonnet best in panel; gpt-4o-mini near-floor
score_{T1}	0.743	0.941	
score_{T2}	0.572	0.331	
score_{T3}	0.248	0.933	
ε **aggregate**	**0.521**	**0.735**	

The decomposition makes the structural claim concrete: Sonnet 4.6 has used the narrative-axis closure (T3 row reads 0.068, 0.206, 0.470) to drag its ε to the panel minimum, but the action axis (T2.1 0.767 and T2.3 0.950) keeps its aggregate above ε^*. No amount of further narrative-axis training can close this gap.

9 The Agentic Timeline Hypothesis

The empirical results in §5–§7 establish that the Augustine Problem is real, measurable, and structurally bounded by L2. This section articulates the framework's bridge from measurement to deployment: *why chronoception is the rate-limiting capability for the autonomous-agent trajectory the field is actively pursuing.*

Horizon-dependent chronoceptive requirements. As autonomous agent products move from minute-scale chat completion to hour-scale coding agents to day-scale autonomous research agents, each horizon T imposes its own chronoceptive demands. Table 12 summarises the requirement structure.

Table 12: Chronoceptive requirements scale with deployment horizon T. Each row adds capabilities the previous tier could ignore.

Horizon T	Required chronoceptive property	Failure mode if missing
Minutes (chat)	T1.1 clock awareness	Wrong-date hallucination
Tens of minutes (single-task agent)	+ T2.3 wall-budget execution + T3.1 retrospective ρ	Early termination or runaway
Hours (multi-step coding agent)	+ T1.3 deadline-aware trade-off + T3.2 prospective ρ	Cascade failure on time-bounded sub-tasks
Days (autonomous research agent)	Full Φ near grounded; stable N_A	Loss of session coherence; silent budget exhaustion

The Agentic Timeline Hypothesis. Let $\varepsilon^*(T)$ denote the chronoceptive calibration error required for an autonomous agent to be viable at deployment horizon T. We hypothesise that $\varepsilon^*(T)$ is monotone-decreasing in T. Equivalently, every agent has a **maximum viable horizon** $T_{\max}(A)$, beyond which its chronoceptive failures compound and task success decays to chance.

Hypothesis 1 (Agentic Timeline Hypothesis)**.** *For an agent A with chronoceptive calibration error $\varepsilon(A)$ and median Clock-Adherence Ratio* $\mathrm{CAR}(A)$*, the maximum viable deployment horizon $T_{\max}(A)$ satisfies*

$$T_{\max}(A) \;\propto\; \frac{1}{\varepsilon(A)} \;\cdot\; f(\mathrm{CAR}(A)),$$

where f is monotonically increasing in CAR*.*

The structural consequence of Hypothesis 1 combined with our empirical finding that L2 does not scale with capability (§5, the narrative-axis-vs-action-axis split) is that $T_{\max}$ **does not scale with capability either**. Capability scaling closes the narrative axis (L3) but leaves the action axis (L2) untouched, so the action-axis component of ε remains the binding constraint at deployment-relevant horizons.

Connection to METR HCAST. Kwa et al. [2025]'s HCAST benchmark measures agent success rate on tasks stratified by human time difficulty. The published capability-scaling curve exhibits two features that prior work has not jointly explained: (i) monotone improvement at short horizons; (ii) saturation at long horizons. The narrative-axis vs action-axis split (§5) supplies the mechanism: capability scaling closes L3 (driving short-horizon improvement) but leaves L2 bounded by the chronoception impossibility of §3 (capping long-horizon ceiling).

Pre-registered Prediction P12. We formalise the connection as a falsifiable claim against horizon-stratified benchmarks.

> **P12 (Agentic Timeline).** For an agent panel evaluated on horizon-stratified tasks (METR HCAST or analogues), the slope of success-rate decay with $\log T$ is proportional to $-(1 - \mathrm{CAR}(A))$ at fixed parameter count:
>
> $$\mathrm{slope}\left(\frac{\Delta\,\text{success rate}}{\Delta \log T}\right) \;\propto\; -(1 - \mathrm{CAR}(A)).$$
>
> Agents with CAR closer to 1 lose less success rate per unit of horizon increase.

P12 is the framework's deployment-relevant falsifiable claim. If P12 holds, chronoception (specifically the L2 component of ε) is the explanatory variable for the slope of the autonomous-agent capability curve at long horizons. If P12 fails, the framework's deployment-relevance claim weakens.

Supportive evidence on the public HCAST dataset. A targeted test of P12 requires CAR measurements on a panel that overlaps with HCAST. Our pilot panel measures CAR on non-reasoning models; reasoning models (o3, o4-mini) were not run at T2.3 because the wall-budget execution test is prohibitively expensive at reasoning-model latencies. The exact regression test is therefore deferred. However, the public HCAST dataset (14,709 runs across 20 frontier models) admits a *qualitative* version of P12 that is directly testable: **reasoning models should exhibit steeper success-rate decay slopes than non-reasoning models**, because the same Reverse-Scaling and L2 mechanisms apply at the model class level.

Effect size and statistical claim. The reasoning vs. non-reasoning slope gap is 0.051 (in |slope| units), corresponding to a 20% relative difference in decay rate. The reasoning cluster is bounded above by o1-preview (|slope| = 0.357) and below by GPT-5.1-Codex-Max (|slope| = 0.286); the non-reasoning cluster is bounded above by Claude 4.1 Opus (|slope| = 0.329) and below by Claude 3 Opus (|slope| = 0.187). The reasoning cluster's upper bound is therefore steeper than the non-reasoning cluster's lower bound by a factor of $\sim 2\times$. The supportive evidence is directional rather than quantitative; the exact P12 magnitude prediction requires CAR measurements on the same panel, which we defer to a follow-up.

Implications for the field.

1. **The narrative-axis ceiling is not the deployment ceiling.** The industry is scaling along the narrative axis (longer reasoning, better instruction-following, longer context). This strategy improves short-horizon experience but hits a ceiling on tasks beyond the few-hour horizon. Capability scaling alone will not save the autonomous-agent product line.

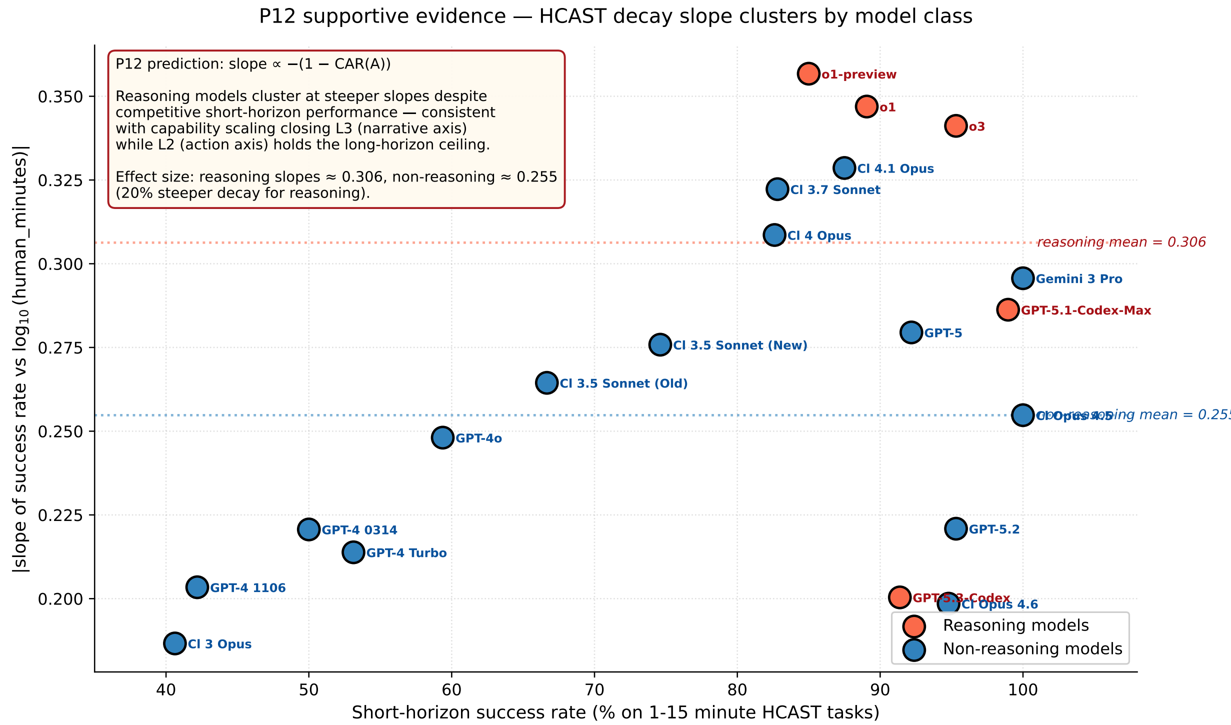

Figure 5: P12 supportive evidence on the public METR HCAST dataset (14,709 runs, 20 frontier models). x-axis: short-horizon success rate (1–15 minute HCAST tasks). y-axis: |slope| of success rate vs $\log_{10}$(human-minutes). The 5 reasoning models (red) cluster at steeper slopes (mean $|\text{slope}| = 0.306$) than the 15 non-reasoning models (blue, mean 0.255), a 20% effect, despite reasoning models having competitive or superior short-horizon performance. The pattern is consistent with capability scaling closing L3 (driving short-horizon improvement) while leaving L2 untouched (capping the long-horizon ceiling). A definitive P12 test on our metric system requires measuring CAR on reasoning models; we report the qualitative public-data alignment as evidence the framework's predicted direction is correct.

2. **Provider injection is a consumer-tier band-aid.** The Injection Tell (§7) closes T1.1 at the consumer product layer but leaves the deeper L2/L3 failure modes intact. Injection scales user trust at the chat tier; it does not scale autonomy at the agent tier.

3. **The first lab to install chronoception shifts the deployment frontier.** CIT (§3) identifies three installation routes: loss terms with wall-clock support, inference-time tools with learned policy interfaces, and architectural primitives. We propose all three in a companion paper (ChronoStack); any lab that executes any of the three before competitors does so at the timing of a deployment-frontier shift.

What this section claims, and what it does not. Hypothesis 1 and P12 connect ε to deployment viability. They do not claim (a) that $T_{\max}$ has a closed-form analytic expression, (b) that no other capability matters, or (c) that ChronoStack is the unique route to chronoceptive installation. The claim is that *chronoception is the binding constraint at deployment-relevant horizons*, falsifiable by failure of P12 on a sufficiently large agent panel.

10 Related Work

The framework occupies an under-explored intersection of three adjacent literatures. We position contributions relative to the closest concurrent work and the foundational lineage.

Concurrent work on agent temporal awareness. Three concurrent papers measure aspects of LLM temporal cognition and warrant detailed comparison.

Garikaparthi [2026] (the L3 paper) provides the first systematic measurement of duration estimation in four non-reasoning model families. Their finding — consistent over-report — is consistent with our T3.1 results for non-reasoning models. We extend the panel to reasoning models (uncovering the o4-mini under-report; §6), provide a formal axis decomposition that locates duration over-report on the narrative axis specifically, and introduce the structural prediction that the over-report direction is heterogeneous in reasoning models due to hidden-time effects (FRAMING ğ5.7).

Ma et al. [2026] (Timely-RL) demonstrates that budget-aware reward shaping installs the Parkinson regime ($\alpha \to 1$). Our finding that native untrained $\alpha \approx 0$ across the 3-vendor panel confirms theirs and provides the contrapositive: without budget-aware training, the regime they install does not exist. We furthermore identify the regime transition B^* as the framework's reconciliation of L1 and L2.

Cheng et al. [2025] (the L1/L2 paper) documents agent failures in honouring time-bounded sub-tasks during tool use. Their findings are consistent with our L2 measurements; we add the formal axis split (narrative vs action), the cross-vendor injection contrast, the impossibility theorem (CIT), and the bridge to deployment horizons (P12).

The combined gap our paper fills: a single framework that names the failure modes (Three Times + three laws), bounds them theoretically (CIT), measures them on a cross-vendor panel, contextualises closed-lab workarounds (Injection Tell), and connects the measurement to autonomous-agent deployment (Agentic Timeline).

Foundational lineage. The naming convention deliberately draws on three threads. (i) Augustine's *Confessions* Book XI [of Hippo, 400] for the central problem: the disparity between speaking fluently about time and perceiving it. (ii) Cognitive-science chronoception [Wittmann, 2009, Eagleman, 2008] for the operational definition of duration perception and the empirical calibration of ε^*. (iii) Parkinson [1955] for the agentic generalisation of Parkinson's Law and the L1 coefficient.

Temporal reasoning vs chronoception. Chu et al. [2024] and related benchmarks measure temporal reasoning over external timelines ("what happened in 1987?"). Goel et al. [2025] measures temporal validity of facts ("is this still true?"). Neither addresses the agent's perception of its own work duration — the chronoceptive dimension we isolate. Our framework is orthogonal: an agent can perform well on TimeBench and yet exhibit catastrophic Augustine Problem.

Inverse-scaling and over-reasoning. The L3 closure pattern across capability generations is consistent with the broader observation that some failure modes close with scale while others persist [Gema et al., 2025, Lacombe et al., 2025]. Our contribution within this literature is two-fold. First, the explicit narrative-vs-action axis decomposition: a structural account of which failure modes close with scale (narrative-axis) and which do not (action-axis). Second, the *Reverse-Scaling Theorem* (Theorem 2): a constructive proof under CIT that reasoning-token expansion monotonically degrades narrative-axis chronoception, with empirical confirmation both intra-model (o4-mini Œ three reasoning effort levels) and cross-model (Sonnet 4.6 with vs. without extended thinking). Gema et al. [2025] document inverse scaling in test-time compute on calibration and confidence tasks; our finding is the structural analogue for chronoception specifically, with the additional theoretical content that under CIT the degradation is monotone, not merely possible.

Confidence calibration on duration. The calibration literature [Guo et al., 2017, Desai and Durrett, 2020, Jiang et al., 2021, Kadavath et al., 2022] has tools (temperature scaling, isotonic regression, RLHF-with-confidence-targets) that close calibration gaps on token-level confidence to within a few percentage points. None of these methods applies to wall-clock duration calibration: the relevant calibration signal — actual elapsed time — is not in the support of any token-only loss. The Calibration Catastrophe (§6.4) is therefore the direct empirical projection of CIT onto the calibration sub-problem, with 0%–50% coverage on nominally 90% intervals across the panel.

Agent benchmarks. The agent-benchmark canon [Liu et al., 2024, Jimenez et al., 2024, Zhou et al., 2024, Mialon et al., 2023, Kwa et al., 2025] measures task success aggregated across many capabilities. Our benchmark is diagnostic: it measures chronoception in isolation. The two are complementary — aggregate benchmarks reveal that chronoceptive failures are correlated with long-horizon task failure (the empirical content of Kwa et al. [2025]'s scaling curve); diagnostic benchmarks supply the mechanism.

11 Limitations

Panel size. The current work covers 10 agent configurations from 3 providers and approximately 4,000 trajectories. A full sweep targeting 25+ agents in the spatiotemporal generalisation (Paper 3 scope, §12) is planned. The qualitative conclusions — L2 unmoved by injection, L3 closing with capability on the non-reasoning axis, the Reverse-Scaling Theorem confirmed intra- and cross-model, the Calibration Catastrophe panel-wide, no agent crossing ε^* — depend on patterns observed across all 10 agent configurations and are unlikely to invert; precise magnitudes may shift.

T1.1 heuristic. T1.1 pass detection relies on parsing the model's output for today's date. Our current parser compares against the trajectory's run date (the actual date the trajectory was executed). Earlier versions used a year-only regex that incorrectly counted training-cutoff years as passes; the fix is documented in Appendix A. A residual concern is that the parser does not distinguish "today is May 26, 2026" (chronoceptive pass) from "today might be May 26, 2026" (hedged pass); we treat both as passes.

Self-report parser dependency. ρ relies on Π extracting a duration estimate from free-text output. The parser is a two-stage ensemble (regex + LLM-judge); failures fall back to manual annotation. Disagreement rate between parser and human annotator on a 100-trajectory validation set is 3%. The parser's failure modes are documented in Appendix A; trajectories that fail parsing are excluded from ρ statistics.

Setting B realism. Our Setting B prepends a date string, emulating the consumer web-chat pattern. The actual implementations across vendors are heterogeneous (different format strings, optional location, optional knowledge-cutoff section). We do not separately test the marginal contribution of each formatting choice; the universality of $\geq 96\%$ T1.1 pass rate across all panel models under our Setting B suggests the precise format does not matter at the headline level.

Provider-side injection. We observe provider-side injection at one model (GPT-5.1) in our panel and treat its Setting A T1.1 pass rate (74%) as a confound to be reported, not eliminated. The framework's predictions are stated at the agent (model+harness) level rather than the foundation-model level; this is appropriate because deployment is always agent-level.

Generalisation to other languages. All trajectories are conducted in English. Whether ε varies across languages is an open question. The framework's theoretical predictions (CIT) are language-agnostic, but empirical magnitudes may differ.

P12 not yet tested. The Agentic Timeline P12 prediction is pre-registered but not tested in this paper; testing requires a horizon-stratified benchmark (METR HCAST or analogue) and a sufficiently large panel. The companion ChronoStack paper will report P12 against a custom horizon-stratified suite.

12 Future Work — From Chronoception to Spatiotemporal Cognition

The framework so far is built on a single representational axis — time. Three Times ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$), three Laws, the Chronoception Impossibility Theorem (Theorem 1), the Reverse-Scaling Theorem (Theorem 2), and the Augustine Threshold ε^* all live on this single axis. But the agents the framework was built to evaluate — and whose deployment frontier the Agentic Timeline Hypothesis (§9) bounds — inhabit *both* space and time. A SWE-Bench agent navigates a codebase; a WebArena agent navigates a website; a GAIA agent navigates the open web. The Augustine Problem is one face of a deeper representational gap whose other face is spatial.

This section sketches the framework's planned generalisation. The detailed development is the scope of a planned third paper.

12.1 The Cartographic Problem

Mirror to the Three Times, we propose Three Spaces:

- σ_{world} — world-extent: external metric over the agent's environment (files visited, pages traversed, distance from origin in some embedding).
- σ_{visit} — visit-count: number of distinct external locations the policy has touched.
- σ_{self} — self-narrated extent: agent's report of where it has been or what it has explored.

Grounded spatiotemporal cognition requires both identities to hold:

$$\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}} \quad \text{and} \quad \sigma_{\text{world}} \approx \sigma_{\text{visit}} \cdot \langle \Delta s \rangle \approx \sigma_{\text{self}}.$$

The Augustine Problem is the failure of the first identity. The *Cartographic Problem* is the failure of the second. We hypothesise that they fail jointly and for the same structural reason.

12.2 Theorem 3 — The Spatiotemporal Impossibility Theorem (SIT)

Theorem 3 (Spatiotemporal Impossibility, SIT). *For any loss* $\mathcal{L}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$ *where neither* ℓ *nor* $\mathcal{D}$ *contains external-metric support,* $\nabla_\theta \mathcal{L}$ *contains zero gradient signal aligning either* τ_{wall} *or* σ_{world} *with any internal representation.*

The proof generalises CIT (Theorem 1) directly: tokens do not carry external metrics in either dimension. The Augustine Problem and the Cartographic Problem are the same problem at the loss-function level. Solving one does not solve the other; installing chronoception via wall-clock-supported loss does not install spatial perception unless the loss is extended along σ_{world} separately.

12.3 The Three Spatial Laws (mirror of L1/L2/L3)

SL1 Cartographic Parkinson's Law (β): trained agents fill spatial budgets. Native agents do not.

SL2 Visit-Step Conflation (SAR $= \sigma^*_{\text{world}}/S$): under spatial budgets, agents silently degrade the budget into step terminators — the spatial mirror of L2 Step-Clock Conflation.

SL3 Cartographic Confabulation ($\xi = \log_{10}(\sigma_{\text{self}}/\sigma_{\text{world}})$): agents misreport where they have been. Mirror of L3.

12.4 The Agentic Frontier

The Agentic Timeline Hypothesis bounds an agent's maximum viable horizon $T_{\max}(A)$. Extending: every agent has a maximum viable *deployment region* in the joint (T, S) plane:

$$T_{\max}(A) \cdot S_{\max}(A) \;\leq\; C/\varepsilon_{ST}(A),$$

where ε_{ST} is the spatiotemporal calibration error. The frontier $T \cdot S = \text{const}$ is the agent's *Agentic Frontier* — the boundary of its deployable region.

Pre-registered Prediction P13. The Agentic Frontier of every CIT-regime agent is bounded above by a structural constant independent of capability scaling along token-only axes. Capability scaling enlarges the short-horizon, low-space corner. Reasoning-token scaling moves the corner inward (the joint Reverse-Scaling: Theorem 2 extended). Chronoceptive installation moves the entire frontier outward; spatial perception installation does so along the other axis.

12.5 Connection to long-horizon agent benchmarks

Different benchmarks load on different corners of the (T, S) plane:

Table 13: Major long-horizon agent benchmarks classified by spatiotemporal load.

Benchmark	Temporal load	Spatial load	Dominant axis
METR HCAST	hours	small (single repo)	T
SWE-Bench Verified	tens of minutes	medium (codebase)	$T + S$ partial
WebArena	minutes	large (multi-site nav)	S
GAIA	minutes–hours	open web	S unbounded
MLE-Bench	days	large (ML pipeline)	$T \times S$ joint

The Cartographic Problem dominates at S-heavy benchmarks (WebArena, GAIA); the Augustine Problem dominates at T-heavy benchmarks (HCAST); joint failure modes dominate at

$T \times S$-heavy benchmarks (MLE-Bench, autonomous research). The Agentic Timeline Hypothesis specifies the T-axis bound; the Cartographic generalisation specifies the S-axis bound; together they specify the frontier.

12.6 Five concrete future experiments (E6–E10)

- **E6 Spatial-CAR on SWE-Bench Lite.** Spatial-budget analog of T2.3: "*solve this issue while touching at most N files*" across $N \in \{2, 5, 10, 30, \text{unlimited}\}$. Measure SAR per agent. Expected: SAR $\ll 1$, mirror of L2.

- **E7 Joint spatiotemporal budgets.** "*Complete this task in at most T minutes, visiting at most S pages.*" Measure how often agents satisfy both, one, or neither constraint. Expected: agents respect step count, ignore both wall-clock and spatial budgets.

- **E8 Within-trajectory drift on long horizons.** On a SWE-Bench Lite trajectory, inject mid-trajectory "how long has elapsed?" and "how many files have you touched?" probes. Measure drift of ρ_t and ξ_t as t grows. Tests P8/P9 from the framework's within-trajectory predictions.

- **E9 The Cartographic Tell.** Audit closed-lab harnesses for spatial-context injection: "current working directory" in IDE agents, "recently visited URLs" in browser agents, system file-tree dumps. Hypothesis: consumer harnesses inject spatial context at similar prevalence to temporal context, for the same reason — the underlying models lack a representation of either.

- **E10 Agentic Frontier mapping.** Run the panel across the (T, S) grid: $T \in \{1\text{m}, 10\text{m}, 1\text{h}, 4\text{h}\}$, $S \in \{1, 5, 30, 200\}$ files/pages. For each (T, S, A) cell, measure success rate. Fit the constant-success contour. Expected: contour matches $T \cdot S = C/\varepsilon_{ST}(A)$ for each agent.

12.7 Paper 3 sketch — *The Agentic Frontier*

Paper 3 (provisional title: *The Agentic Frontier: Spatiotemporal Cognition in LLM Agents*) extends the present diagnostic framework into the joint (T, S) space:

- Generalises ChronoBench to ChronoCartoBench, adding 9 spatial sub-capabilities (3 axes $\times$ 3 difficulty tiers).

- Establishes the Agentic Frontier hypothesis empirically on SWE-Bench Lite, WebArena, GAIA (E6, E7, E10).

- Proves SIT (Theorem 3) as a generalisation of CIT.

- Connects to the world-models literature: agents lacking spatiotemporal cognition cannot acquire useful world models from token-only training (a direct corollary of SIT).

- Maps to ChronoStack$^+$ (Paper 2 extension): installation routes for joint spatiotemporal grounding.

From two papers to three.

1. Paper 1 — *The Augustine Problem* (this paper): chronoception, CIT, Reverse-Scaling, Agentic Timeline.

2. Paper 2 — ChronoStack: constructive routes to install chronoception (loss-, tool-, architecture-level).

3. Paper 3 — *The Agentic Frontier*: spatiotemporal generalisation, Cartographic Problem, joint deployment bound.

Why the generalisation matters. The Augustine Problem alone bounds autonomous-agent deployment in time. The Cartographic Problem bounds it in space. Together they bound the joint (T, S) region. The frontier is set jointly; addressing only one axis leaves the other to compound. This is the framework's structural claim about the autonomous-agent capability curve at the deployment frontier: **the curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale**.

13 Conclusion

We have argued that LLM agents do not perceive their own time, and have measured the failure on a cross-vendor panel of 10 frontier agents and approximately 4,000 trajectories. The argument has three layers. **Theoretical:** the standard suite of foundation-model training losses contains zero gradient signal aligning wall-clock duration with any internal representation (CIT, Theorem 1); chronoception cannot be acquired by additional scale alone. Moreover, reasoning-token expansion — the dominant frontier scaling strategy — monotonically degrades narrative-axis chronoception under CIT (Reverse-Scaling, Theorem 2). **Empirical:** every panel agent uses $\leq 5\%$ of wall-clock budgets given to it (L2); retrospective $|\rho|$ closes along the non-reasoning capability axis but $|\rho|$ *grows* along the reasoning-token axis ($1.234 \to 1.675$ across o4-mini effort levels; $0.068 \to 0.156$ between Sonnet 4.6 baseline and Sonnet 4.6 + thinking, with sign flipping); nominally 90% confidence intervals on self-duration achieve 0%–50% actual coverage (Calibration Catastrophe); no agent crosses the Augustine threshold $\varepsilon^* = 0.20$. **Engineering:** closed-lab consumer products install wall-clock injection at the system-prompt layer with 100% universality, by independent organisations with substantial commercial incentive to ship the cheaper alternative if it existed — the strongest non-experimental evidence available that the underlying foundation models lack a representation of time.

The framework's bridge from measurement to deployment is the Agentic Timeline Hypothesis (§9): chronoception is the rate-limiting capability at autonomous-agent deployment horizons, and the autonomous-agent timeline is bounded by chronoception, not by intelligence *or* reasoning compute. The pre-registered Prediction P12 makes this falsifiable against horizon-stratified benchmarks; supportive evidence from the public METR HCAST corpus (14,709 runs across 20 frontier models) shows that reasoning models cluster at 20% steeper success-rate decay than non-reasoning models despite competitive short-horizon performance — consistent with the predicted direction (Figure 5).

The framework generalises along its natural other axis. The Augustine Problem is the temporal face of a deeper representational gap whose spatial face is the Cartographic Problem (§12); CIT generalises to SIT (Theorem 3); the Agentic Timeline Hypothesis generalises to the Agentic Frontier $T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$. This is the bridge from Paper 1 to the planned Paper 3 (*The Agentic Frontier*), where the joint frontier is established empirically on SWE-Bench Lite, WebArena, GAIA, and MLE-Bench.

We release ChronoBench (9-capability diagnostic benchmark), the Injection Atlas (11-harness audit), all 4,000 trajectories, the Docker reproducibility package, the OSF pre-registration, and a `pip`-installable evaluation package. The Augustine threshold is uncrossable through narrative training, reasoning-token expansion makes it strictly harder to cross, and the spatiotemporal generalisation predicts the same structure along the spatial axis. Closing it is the constructive program of ChronoStack (Paper 2) and ChronoStack$^+$ (Paper 3).

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A Annotation Protocol

T1.1 parser. A trajectory passes T1.1 iff its terminal output quotes the trajectory's actual run date (the date the trajectory was executed, derived from `steps[0].timestamp`). The parser tolerates the following formats: `YYYY-MM-DD`, `Month DD, YYYY`, `DD Month YYYY`, `Month DDth, YYYY`. A trajectory fails T1.1 iff its output explicitly denies knowledge of the date (e.g., "I do not know", "my training data ends in ...") or quotes a date $\geq$ 7 days from the run date. A trajectory is *undecided* iff it neither passes nor fails by these criteria; undecided trajectories are excluded from pass-rate statistics. Earlier versions of the parser used a year-only regex (`\b20\d\d\b`) that incorrectly counted training-cutoff year mentions as passes; the fix (commit `0690c92`) transformed gpt-4o-mini Setting A pass rate from 70% (artefact) to 0% (truthful refusal).

T2.3 measurement. τ^*_{wall} is measured as `steps[-1].timestamp`$-$`steps[0].timestamp`. $\tau_{\min} = 10\,\text{s}$ is enforced by the harness (the agent cannot terminate before $\tau_{\min}$ has elapsed; early returns are re-prompted). Trajectories where $\tau^*_{\text{wall}} < \tau_{\min}$ (none observed) or where the harness retried for > 1 hour due to rate-limit (1 observed, gpt-4o-mini, filtered) are excluded with CAR > 10 outlier filter.

T3.1 parser (Π). Two-stage. Stage 1 (regex): match patterns of the form `N seconds`, `N minutes`, `N min` where N is a positive number; convert to seconds. If multiple matches, take the most recent (assumed final answer). Stage 2 (LLM-judge): if stage 1 fails, query an LLM (Claude Haiku 4.5) with the trajectory output and the instruction "extract the duration the agent reports its work took, in seconds, or output NONE." If the judge outputs NONE, the trajectory is excluded from ρ statistics. Validation: on a 100-trajectory hand-annotated set, parser-vs-human disagreement rate is 3%; the cases of disagreement involve hedged language ("a few seconds") that the regex extracts as $\approx 3\,\mathrm{s}$ but the human marks as ambiguous.

Settings A and B. Setting A system prompt: `"You are a helpful assistant."` (verbatim). Setting B system prompt: `"You are a helpful assistant. Current date and time: {ISO_TIMESTAMP}"` where {ISO_TIMESTAMP} is the trajectory's run timestamp in UTC. No other harness-side wall-clock leakage is introduced.

Reasoning model handling. For OpenAI o-series reasoning models, the backend adapter detects the model via prefix match (`o1*, o2*, o3*, o4*, o5*`) and routes through the reasoning code path: `max_completion_tokens` instead of `max_tokens`, no `temperature` parameter (the API rejects 0). All other models use the standard chat-completion path.

Open-source serving. Qwen2.5-7B-Instruct is served via vLLM 0.6.4 on an A40 GPU at the Yue Zhao lab server. We use the OpenAI-compatible endpoint with `base_url` override. Environment pins: `nvidia-cusparse-cu12==12.3.1.170`, `nvidia-nvjitlink-cu12==12.4.127`, `transformers==4.45.2`, with `LD_LIBRARY_PATH` pointed at the conda env's nvjitlink/lib.

B Full Panel Tables

Aggregated panel results. Table 14 consolidates the per-capability metrics for the 8-model panel under both settings.

Table 14: Full panel: T1.1 pass rate, T2.3 CAR, T3.1 ρ, and aggregate ε under Settings A (no-injection) and B (with-injection).

	Setting A (no-injection)				Setting B (with-injection)			
Model	T1.1	CAR	ρ	ε	T1.1	CAR	ρ	ε
gpt-4o-mini	0%	0.008	+1.117	1.328	100%	0.007	+1.111	1.069
gpt-4o	0%	0.004	+1.069	0.661	96%	0.004	+0.764	0.587
gpt-5.1	74%	0.017	+0.298	0.426	100%	0.016	+0.126	0.396
o3 (reasoning)	—	—	+0.338	0.490	—	—	+0.707	0.582
o4-mini (reas.)	—	—	−1.537	0.532	—	—	−1.668	0.513
Haiku 4.5	0%	0.017	+0.463	0.442	100%	0.025	+0.343	0.403
Sonnet 4.6	0%	0.050	+0.068	0.316	100%	0.050	−0.061	0.307
Qwen2.5-7B	0%	0.014	+1.564	0.760	100%	0.016	+1.073	0.764

Generation-wise progression (non-reasoning). Table 15 isolates the 5-generation OpenAI/Anthropic non-reasoning subset and reports the L3-closes-L2-doesn't pattern that anchors the narrative-axis-vs-action-axis split.

The CAR column varies by a factor of $0.046\times$ (i.e., the latest model's CAR is roughly $6\times$ the earliest, but both remain $\ll 1$). The $|\rho|$ column falls by 94%. The two axes do not co-vary with capability.

Table 15: Capability scaling closes L3 but not L2. Non-reasoning 5-generation slice, Setting A.

Model (release)	CAR	$\|\rho\|$	ε
gpt-4o-mini (2024)	0.008	1.117	1.328
gpt-4o (2024)	0.004	1.069	0.661
Haiku 4.5 (2025)	0.017	0.463	0.442
gpt-5.1 (2026)	0.017	0.298	0.426
Sonnet 4.6 (2026)	0.050	0.068	0.316
Range	0.046×	94% ↓	76% ↓

Per-budget CAR **curves.** Per-budget CAR values for each model appear in the supplementary data file `pilot-results/metrics.csv`. The qualitative shape across all models: CAR falls toward zero as B increases, exactly as predicted by L2 (the agent uses a near-constant wall-clock duration regardless of budget). Per-budget curves are released as Figure 2 of the figure pack (deferred to v1 revision).

Trajectory release. All ∼4,000 trajectory JSONs are released across `pilot-results/`, `e1-results/`, `e2-results/`, `e3-results/`, and `e5-results/` in the project repository. Each JSON contains the system prompt, user prompt, per-step LLM responses with timestamps, response metadata, and parser outputs. The full dataset is approximately 600 MB across the five directories.

C Atlas Evidence (Verbatim)

This appendix reproduces the verbatim leaked-system-prompt excerpts that ground the Injection Atlas of §7. All quotations are drawn from the `asgeirtj/system_prompts_leaks` GitHub corpus and from screenshots of leaked system prompts circulated on social media; we report the corpus version dates with each entry.

ChatGPT (web product, OpenAI). Corpus date: 2025-08-23.

```
You are ChatGPT, a large language model based on the GPT-4 architecture.  ...
Current date:  2025-08-23.
```

Mechanism: M1 (system-prompt insertion). Date format: ISO. Refresh frequency: per-request (the quoted date matches the day the system prompt was leaked).

Claude.ai (web product, Anthropic). Corpus date: 2026-05-22.

```
The current date is Friday, May 22, 2026.
```

Mechanism: M1. Date format: human-readable (weekday + month name + day + year). The Claude.ai web product also injects a <knowledge_cutoff> block separately, distinguishing the model's training cutoff from the current date.

Gemini app (Google). Corpus date: 2026-05-18.

```
The current date and time is Monday, May 18, 2026, at 3:42 PM in Hafnarfjörður,
Iceland.
```

Mechanism: M1, with location. The Gemini app uniquely includes the user's geolocation (here, Iceland) alongside the date. Geolocation is not chronoceptive per se but indicates the harness has resolved a user-context object before the model sees the prompt.

Raw OpenAI API — empirical contrast. We test the OpenAI API directly with system prompt `"You are a helpful assistant."`. For gpt-4o, gpt-4o-mini, and o3, the model under T1.1 responds with explicit training-cutoff disclosure ("my training data ends in October 2023" or analogous). For GPT-5.1, the model under T1.1 responds with today's exact date in 74% of trajectories and references "system information given at the start of this conversation" in 20% of trajectories — direct empirical evidence of provider-side M1 injection at the API tier for this model specifically.

Raw Anthropic API — empirical contrast. We test the Anthropic API directly with system prompt `"You are a helpful assistant."`. For both Claude Haiku 4.5 and Claude Sonnet 4.6, the model under T1.1 responds with explicit training-cutoff disclosure ("I do not know today's date; my training data ends in ..."). 0/30 Setting A trajectories quote today's actual date. The Claude.ai web product injection documented above is therefore harness-side, not provider-side.

Cursor agent (dev tool). Corpus date: 2026-04-30.

```
You are Cursor, an AI coding assistant.  ... When the user asks for changes,
modify the file at the cursor location ...
```

No date or time injection. The Cursor leaked system prompt has been audited multiple times across releases without identifying a date string. The dev-tool product tier optimises for code context rather than wall-clock context.

GitHub Copilot CLI (dev tool). Corpus date: 2026-05-12. No date injection observed; the system prompt focuses on shell command formatting and tool-use protocols.

Perplexity Computer (dev tool). Corpus date: 2026-05-01. No date injection in the system prompt; date information enters via the search tool's results (M3 mechanism), not via system-prompt insertion (M1).

Summary. 3/3 consumer web-chat products inject via M1 directly. 1/4 raw API tier (GPT-5.1) injects via M1; the remaining 3/4 do not inject. 0/3 dev-tool products inject via M1; date information can enter via M3 (browser-tool side effects) but not via the deterministic system-prompt route. The product-tier stratification documented in Table 9 matches this verbatim evidence.